Akka vs. Google Gemini Enterprise Agent Platform

A comparison for teams building agentic AI

October 2026

The Gemini Enterprise Agent Platform is a set of Google Cloud services the customer integrates and operates. At Cloud Next 2026, Google rebranded Vertex AI as the Gemini Enterprise Agent Platform — a code-first kit (ADK), a low-code builder (Agent Studio), and a managed runtime (Agent Runtime, formerly Agent Engine) under a new name. The capability is real. The agent runtime is excluded from the platform SLA. Each piece bills on its own meter as load grows, and identity, models, and data are bound to Google Cloud. The Akka Agentic AI Platform delivers agents, memory, streaming, APIs, and governance as one platform that guarantees resilience and scalability — at six-nines availability, backed by indemnities.

99.5%	99.9999%	4ms	90%
GEMINI ENT. INFERENCE SLA	AKKA AGENTIC AI PLATFORM SLA	AKKA STATE READS	LOWER INFRA COST ON AKKA

DIMENSION	GEMINI ENTERPRISE AGENT PLATFORM	AKKA
What it is	A rebrand of Vertex AI (Cloud Next 2026): a set of Google Cloud services — ADK, Agent Studio, Agent Runtime, Agent/Model Garden — for building agents	The Akka Agentic AI Platform — a platform that guarantees resilience and scalability
Scope	Build tools plus a managed agent runtime; memory, governance enforcement, and the API/streaming tier are assembled from separate Google Cloud services	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.5% online inference (Gemini API; custom models on 2+ nodes); 99.5% Pipelines; the Agent Runtime is excluded from the SLA	99.9999% — entire platform, backed by indemnities
RTO / RPO	Per underlying Google Cloud service; customer-architected across regions	Sub-1-minute RTO; zero-byte RPO; active-active
Naming / churn	Renamed repeatedly: Enterprise Search → Generative AI App Builder → Vertex AI Search & Conversation → AI Applications → Agent Builder → Gemini Enterprise Agent Platform	Stable platform; since 2009; 100,000+ deployments
Governance / EU AI Act	IAM, VPC-SC, CMEK, DLP, audit logs, data residency; Agent Gateway access policies, Model Armor, and semantic governance policies (preview) for agents on Agent Runtime; no EU AI Act classification	Aspect-woven runtime enforcement + full pre-production governance
Lock-in	Google Cloud IAM identity, GCP-project data residency, models served through GCP	Akka cloud, any hyperscaler VPC, own Kubernetes, on-prem, or sovereign cloud; portable specs
Cost model	Pay-as-you-go: $0.085/vCPU-hour + $0.009/GiB-hour runtime, $0.30/GiB-month memory and session storage plus operations, per-token model calls, per-query search	Shared compute; up to 90% lower infrastructure for the same workload; fixed annual fee
Certifications	Google Cloud compliance program	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

A Collection of Pieces Under One Console

The Gemini Enterprise Agent Platform is the new name for Vertex AI, announced at Google Cloud Next 2026 on April 22, 2026. The platform carries a code-first **Agent Development Kit (ADK)**, a low-code **Agent Studio** canvas, and prebuilt templates in **Agent Garden**, plus a managed **Agent Runtime** (formerly Agent Engine) and a **Model Garden** of 200+ models (Gemini and Anthropic Claude among them). Each is a separate Google Cloud service the customer wires together and operates; the platform unifies the console and the namespace; the runtime underneath stays a set of services.

Capability	Gemini Enterprise Agent Platform	Akka
Agent build tools	ADK + Agent Studio + Agent Garden	Built in
Managed agent runtime	Agent Runtime — excluded from the SLA	Built in, covered by the platform SLA
Durable memory	Memory Bank, metered per GiB-month and per operation	Built in, 4ms / sub-10ms
Real-time streaming	Provisioned from separate Google Cloud services	Built in, backpressured, petabyte-scale
Governance / policy enforcement	Agent Gateway access policies, Model Armor, semantic governance policies (preview); agents on Agent Runtime	Inline, runtime-embedded
Pre-production governance	Agent Simulation and Agent Evaluation; no classification against a named regulation	Classification, sign-offs, sealed posture

Akka delivers all of it as one runtime with one SLA. The customer does not integrate or operate the seams between an orchestration kit, a memory store, a streaming tier, and a governance stack.

Rebrand and Naming Churn

The product the customer would standardize on has been renamed repeatedly: Enterprise Search → Generative AI App Builder → Vertex AI Search & Conversation → AI Applications → Agent Builder → and now, at Cloud Next 2026, the Gemini Enterprise Agent Platform. The rebrand is additive — existing Vertex AI workloads run unchanged under the new namespace, and SDKs, billing, and APIs were migrated with no breaking changes. But the churn is real and recent: the `vertexai.generative_models` Python SDK was deprecated June 24, 2025, with removal scheduled for June 24, 2026 — a live migration deadline. Akka has been one stable platform since 2009 and 100,000+ production deployments; the name on the contract has not changed under the customer.

Availability and Disaster Recovery

Google Cloud publishes a 99.5% monthly-uptime SLA for the Gemini online inference API, and 99.5% for online prediction of custom models deployed across 2 or more nodes, which the customer must architect and pay for. Vertex Pipelines carry 99.5% and the training-cluster control plane 99%. Most consequentially for agents: **the Agent Runtime (formerly Agent Engine) is explicitly excluded from the Gemini Enterprise SLA**, as are user-defined agents created in the Gemini Enterprise environment. Akka publishes a 99.9999% availability SLA across the entire platform — agents, memory, streaming, and governance — with sub-1-minute RTO, zero-byte RPO, active-active across regions, and contractual indemnities.

Metric	Gemini Enterprise Agent Platform	Akka
Availability SLA	99.5% online inference (2+ nodes)	99.9999%
Allowed downtime / year	~43.8 hours	~31 seconds
Agent runtime SLA	Excluded from the SLA	Covered
RTO / RPO	Per underlying service; customer-architected	Sub-1-minute / zero-byte
SLA scope	Individual Google Cloud services	The entire platform

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume. List prices are a separate question. The Gemini Enterprise Agent Platform bills **pay-as-you-go** on meters that scale with load: the Agent Runtime at $0.085 per vCPU-hour and $0.009 per GiB-hour, session and memory storage at $0.30 per GiB-month plus read and write operations, Agent Gateway calls, model calls per token, and search at $1.50 per 1,000 queries — with per-seat subscriptions on the agent app layer on top.

Akka runs orchestration, agents, memory, streaming, APIs, observability, and governance on one shared-compute runtime. The efficiency comes from actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing. The spend is a predictable fixed annual fee.

Google Cloud Lock-In

The platform is bound to Google Cloud. Identity runs through Google Cloud **IAM** (or a federated IdP into it); data residency is scoped to a Google Cloud project and region (DRZ); and the Model Garden's models are served through GCP. Portability across this stack is the customer's project.

Akka deploys on Akka cloud, any hyperscaler VPC (AWS / Azure / GCP), the customer's own Kubernetes, on-prem, or a sovereign cloud with country-isolated data, networking, and local SREs — from one portable spec. The platform you standardize on is not the platform you are confined to.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. The loop runs on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model choice** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Governance and the EU AI Act

The Gemini Enterprise Agent Platform provides infrastructure governance: Google Cloud IAM, VPC Service Controls, Customer-Managed Encryption Keys, Data Loss Prevention, Access Transparency, audit logging, and data residency. For agents on Agent Runtime, Agent Gateway blocks every connection that no IAM access policy allows, and Model Armor on the gateway blocks prompt injection, jailbreaks, and sensitive-data leaks. Semantic governance policies, in preview since June 2026, check each proposed tool call against plain-language rules and deny calls that break them, with a rationale. Agent Gateway currently covers agents on Agent Runtime, with Cloud Run, GKE, and other clouds listed as coming soon. Compliance Manager offers Google's recommended AI security practices and NIST AI 600-1 privacy controls (preview) as its AI frameworks. Neither framework maps to the EU AI Act, and the platform offers no pre-deployment classification against a regulatory corpus.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

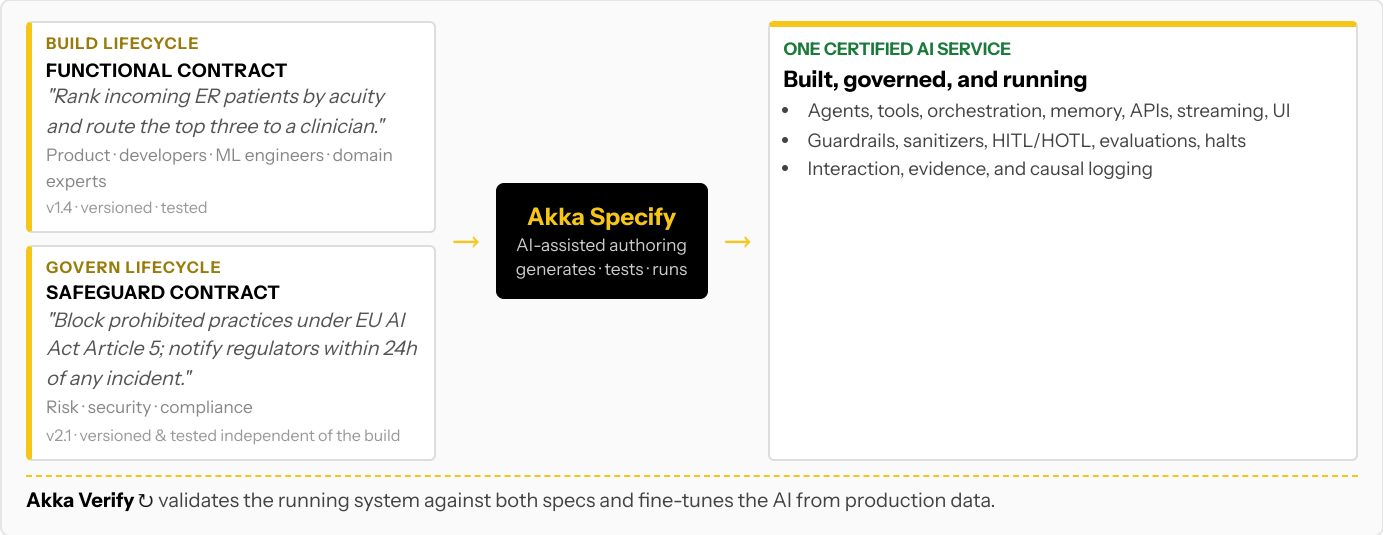
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; classification against 241 regulations and frameworks and 1,718 controls (699 of which carry a financial penalty) before a system ships; multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Akka Verify applies any control to any AI system and guarantees its enforcement.

Build and Governance Run as Independent Lifecycles

Building on the Gemini Enterprise Agent Platform means engineers in ADK or builders in Agent Studio producing agent logic; a separate team writes Agent Gateway access policies and semantic governance policies that apply to those agents on Agent Runtime. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences. Google's spec tooling is Conductor, a Gemini CLI extension in preview that keeps specifications and plans as Markdown files in the repository.

Real-Time Streaming at Petabyte Scale

The Gemini Enterprise Agent Platform has no streaming engine; real-time pipelines are provisioned from separate Google Cloud services and wired to the agents. Akka streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Fox's real-time personalization engine at 5 billion tokens per second).

For the Buyer: Maturity, Adoption, and Accountability

Google Cloud is durable; that is not the question. The question is the **product's** maturity and the integration burden it puts on the buyer.

Buyer concern	Gemini Enterprise Agent Platform	Akka
Product maturity	Rebranded at Cloud Next 2026 (April); GA core, with agent components actively evolving; the prior SDK is being removed June 2026	Stable platform; since 2009; 100,000+ deployments (52 banks)
Naming / churn	Renamed five times (Enterprise Search → Agent Builder → Gemini Enterprise Agent Platform)	One platform, one name under the customer
Scope of accountability	The buyer integrates and operates ADK, Agent Runtime, memory, streaming, and governance as separate Google Cloud services	One platform, one SLA, 24/7 SRE — Akka owns the running system
Availability commitment	99.5% inference; Agent Runtime excluded from the SLA	99.9999% across the platform, with indemnities
Portability / lock-in	Google Cloud IAM identity, GCP-project data, models served through GCP	Akka cloud, any hyperscaler VPC, own Kubernetes, on-prem, or sovereign cloud; portable specs
Certifications & audits	Google Cloud compliance program	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Budget predictability	Pay-as-you-go meters that scale with load	Fixed annual fee finance can forecast

The decision is scope, accountability, and portability: the Gemini Enterprise Agent Platform gives the buyer a strong set of Google Cloud building blocks to assemble inside Google Cloud; Akka gives the buyer one governed platform, with one SLA, that runs anywhere.

Customers Running Agentic and Real-Time Systems on Akka

Manulife	Fox	Swiggy	John Deere	Verizon
2,000	**150k→22k**	**144→71ms**	**1,000+**	**750%**

developers in 6 countries under central risk control	cores after porting the AI personalization engine to Akka	prediction latency, at 22% fewer tokens	tractor sensors turned into real-time insight	order-processing capacity gain; 6s → 2.4s response

Common Questions

We are already on Google Cloud and starting with the Gemini Enterprise Agent Platform. Why add Akka?

The platform gives you good building blocks — ADK, Agent Studio, Agent Runtime, Model Garden. A production agentic system also needs durable memory, streaming, an API tier, and inline runtime governance unified under one SLA. On Google Cloud you integrate those from separate services and own the seams; the Agent Runtime itself is excluded from the SLA. Akka delivers them as one platform at 99.9999%, and runs in your Google Cloud VPC if you want to stay there.

The rebrand kept existing workloads running. Doesn't that mean the platform is stable?

The rebrand is additive and existing workloads run unchanged. But the agent layer has been renamed five times in a few years, and the prior generative-models SDK is being removed in June 2026. Enterprise buyers weigh that churn when they standardize. Akka has run as one platform under the same name since 2009 and 100,000+ deployments.

Can we govern for the EU AI Act with the platform's security controls?

You get IAM, VPC-SC, CMEK, DLP, audit logs, and data residency, plus Agent Gateway access policies, Model Armor, and semantic governance policies (preview) for agents on Agent Runtime. No Compliance Manager framework on the platform maps to the EU AI Act. The EU AI Act also expects immutable records witnessed as they happen, human override of running agents, pre-deployment classification, and a sealed audit artifact. Akka embeds all of this and classifies a system against 241 regulations and frameworks and 1,718 controls before it ships.

Isn't pay-as-you-go cheaper than a platform fee?

Pay-as-you-go means several meters that scale with load — runtime vCPU/GiB-hours, memory and session storage and operations, gateway calls, per-token model calls, per-query search — plus per-seat subscriptions on the app layer. The Akka shared-compute model is up to 90% cheaper to operate for the same agentic transaction volume, on a fixed annual fee finance can forecast.

Sources

Rebrand / Cloud Next 2026: cloud.google.com/blog/products/ai-machine-learning/introducing-gemini-enterprise-agent-platform — Vertex AI rebranded April 22, 2026; existing workloads run unchanged

Components: docs.cloud.google.com/gemini-enterprise-agent-platform/overview, /agent-studio/design-agents, /build/runtime; cloud.google.com/model-garden — ADK, Agent Studio, Agent Runtime (formerly Agent Engine), Agent/Model Garden; 200+ models incl. Gemini and Anthropic Claude

Availability SLA: cloud.google.com/vertex-ai/sla, /vertex-ai/generative-ai/sla — 99.5% Gemini online inference API (modified 2026-06-30); 99.5% custom-model online prediction on 2+ nodes, 99.5% Pipelines, 99% training control plane (modified 2026-02-12)

Agent runtime SLA exclusion: cloud.google.com/terms/gemini-enterprise/sla — SLA does not apply to user-defined agents or to agents via Agent Runtime on the Gemini Enterprise Agent Platform

Pricing (pay-as-you-go): cloud.google.com/products/gemini-enterprise-agent-platform/pricing — Agent Runtime $0.085/vCPU-hour + $0.009/GiB-hour; Memory Bank and Sessions $0.30/GiB-month plus operations (from 2026-09-01); Agent Gateway billed from 2026-07-13; Vertex AI Search $1.50 per 1,000 queries

Naming churn / SDK deprecation: docs.cloud.google.com/vertex-ai/generative-ai/docs/release-notes — `vertexai.generative_models` deprecated June 24 2025, removal June 24 2026; prior names incl. Enterprise Search, Generative AI App Builder, Agent Builder

Governance / lock-in: docs.cloud.google.com/gemini-enterprise-agent-platform/machine-learning/general/vpc-service-controls; /build/runtime — IAM, VPC-SC, CMEK, DLP, Access Transparency, audit logging, data residency (DRZ); docs.cloud.google.com/gemini-enterprise-agent-platform/govern/gateways/agent-gateway-overview; docs.cloud.google.com/model-armor/model-armor-agent-gateway-integration; docs.cloud.google.com/gemini-enterprise-agent-platform/govern/policies/semantic-governance-overview (preview); docs.cloud.google.com/security-command-center/docs/compliance-manager-frameworks

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 241 regulations and frameworks / 1,718 controls / 699 with a financial penalty; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital